

Model Card — AOT-GOAT (Recipient-Aware Message Adaptation)

Build–Audit–Pitch mini NLP project · Research prototype · Synthetic evaluation · Figures from the final runs (aot-goat.zip, github.com/painsparc/AOT-GOAT)

1. Model details

Model name / task	AOT-GOAT (AOT = decision gate, GOAT = minimal-intervention selection). Detects recipient-conditioned ambiguity in a draft message, given the sender–recipient history, and only if needed makes the smallest evidence-grounded edit. Risk types: coreference, temporal, scope, terminology, implicit context.
Architecture	Deterministic rule / lexicon / heuristic pipeline in Python (not a trained neural model; nothing is trained): history → dyadic profile → rule-based risky spans → risk score → AOT pass/intervene → GOAT greedy edit set → template replacement → hard-constraint check. Optional LLM adapter (Anthropic / OpenAI-compatible) is off in all reported results. $\text{risk} = \text{base} \times (1 - \text{grounding}) \times (1 + 0.5 \times \text{clarification_rate})$.
Thresholds (CFG)	τ_{span} 0.35 (chosen on the seed-1 calibration set, then frozen) · τ_{high} 0.55 · τ_{low} 0.30 · noise floor 0.25 · residual 0.15 · max 14 added tokens per edit. Intervene if any $\text{span} \geq \tau_{\text{span}}$ or message risk ≥ 0.55 .
Preprocessing	History “S: ... / R: ...” strings normalised; lowercase regex tokenisation [a-z0-9]+; stop-word and common-acronym lists; regex lexicons (dates, jargon, scope, anchor phrases); time-zone parsing (UTC±h, IST/EST/PST...). Software: Python 3.9+ (developed on 3.12), flask, requests, python-dotenv, gunicorn; no ML libraries.
Developed by	painsparc (GitHub) · NLP course project, October 2026.

2. Data used (no training data; no human-annotated data)

Dataset	Size	Origin	Used for
Controlled set, dev	41 cases	Hand-written	Building / debugging rules
Controlled set, holdout	18 cases	Hand-written after rules were done	Checking only, never tuned on
Calibration set (seed 1)	720 cases, 120 simulated dyads	Seeded generator	Choosing the threshold; bug finding
Test set (seed 2)	1,800 cases, 300 dyads: single 900, mixed 300, hard_pos 300, hard_neg 300	Seeded generator	Reported results, scored once, threshold frozen
Demo scenarios (sample_data.json)	2 messages, 5 recipient histories	Written for the UI demo	Demo only
MultiWOZ 2.2, one shard (dialogues_001.json, MIT; verify before citing)	2,000 real system turns, fixed seed, ≤ 8 earlier turns as history	Real, unlabeled ; fetched by python evaluate.py --real (not in the ZIP)	Robustness check only: no training, tuning, calibration or accuracy

Gold labels for generated cases come from how each case was built (was the risky span grounded in the history), never from pipeline output. Gold-positive rate on the test set: 12.6% of (case, risk type) pairs. **Real labeled conversations: 0**. Every accuracy figure is from synthetic or hand-written data.

3. Intended use

- Intended:** research / educational prototype that helps a sender writing to a recipient who lacks shared context (e.g., a new contact in another time zone) by flagging ambiguous spans and applying minimal edits.
- Out of scope:** production communication-safety use; non-English text; unmarked or paraphrased implicit context; any claim of real-world accuracy or fairness; public deployment with an LLM key (no authentication or rate limiting).

4. Performance (offline analyzer, risk-type level, micro-averaged over (case, risk type) labels)

Evaluation	Precision	Recall	F1 (95% CI)
Test seed 2, designed scenarios (single + mixed, 1,200 cases)	1.000	0.998	0.999 (0.997–1.000)
same detector, recipient profile switched off	0.571	1.000	0.727 (0.707–0.747)
Test seed 2, all 1,800 cases incl. hard slices	0.830	0.774	0.801 (0.787–0.816)
same detector, recipient profile switched off	0.522	0.776	0.624 (0.604–0.645)
Controlled set: dev (41) / holdout (18) / overall (59)	0.929 / 1.000 / 0.951	1.000 / 0.929 / 0.975	0.963 / 0.963 / 0.963
Controlled set, decision level PASS vs INTERVENE (59)	0.969	0.969	0.969

Recipient conditioning adds 0.25–0.29 F1 over the identical detector without history (95% CI, resampling dyads). Controlled-set decision counts: TP 31, TN 26, FP 1, FN 1 (FPR 3.7%, FNR 3.1%). **Safety checks:** all 610 adapted outputs passed intent, no-invented-facts and minimal-edit checks; an independent check against generator facts found 0 untraceable tokens and 0 polarity changes (checker is a gate, so this is partly by construction). 28 end-to-end tests (LLM mocked).

5. Performance by slice, real-text check and sample predictions

- Hard slices (seed 2):** hard_pos (risky wording outside the rule vocabulary, e.g. “what we did for Mumbai”, lowercase acronyms) recall is only **0.153**; on hard_neg (idioms such as “Welcome to the team!”) every flag is a false positive (104 of 1,500 type decisions).
- Real text (MultiWOZ, unlabeled):** 0 crashes and 0 constraint violations on 2,000 turns. Using real history lowered flag rates: coreference 21.2% → 13.6%, temporal 8.6% → 5.5%, scope 20.1% → 16.7%. It exposed one checker false alarm (messages without final punctuation), since fixed. These are flag rates, not accuracy.

Demo message: “Please send it to everyone by EOD tomorrow, as discussed. The SLA needs sign-off before the meeting.”

Recipient (history)	Decision	Output
Asha — same project and time zone, has seen all of it	PASS	Message returned unchanged.
Vikram — new contact, 10.5 h time-zone gap, nothing shared	INTERVENE	All six detected spans edited using sender context, history and computed dates.
Meera — knows the report and the EOD convention, not “everyone” or “SLA”	INTERVENE	Only <i>it</i> , <i>everyone</i> , <i>SLA</i> edited; rest kept verbatim.
“Please share your feedback soon.” (neutral history)	Needs sender input	“soon” has no supporting evidence: left as written and the sender is asked for an exact date/time.

6. Explainability

Intrinsic, pipeline-level: each output lists every span with its **type, reason, base severity, grounding score and note (history evidence), clarification rate, risk, AOT decision, GOAT selection and the edit with its evidence**, so each change is traceable. No post-hoc neural explainers (SHAP, LIME, attention) are used or claimed.

7. Limitations and ethical considerations

- Evaluation validity:** the generator and the rules share an author and a view of what “grounded” means, so the near-perfect designed-scenario score shows consistency and the benefit of the profile, **not real-world accuracy**. Bugs found on the calibration seed were fixed before the test seed was scored; the real-text checker fix came later and affects safety checks only.
- Not safe to claim:** any accuracy on real conversations or validation on a real labeled dataset. **Missing step:** independent human annotation of real conversations and human ratings of “same intent / no new facts”.
- Coverage:** English only; lexicon / regex based; weak recall on unseen wording (0.153); idiom false positives; sender-supplied context is trusted as given. LLM mode was tested only with mocked HTTP and is not bit-for-bit reproducible.
- Fairness:** no demographic or sensitive attributes in any data, so no demographic fairness analysis; no claim of being unbiased or fair. It does **not** implement RLM, IDT, Cox model, APD, embeddings or fine-tuning.

8. Reproducibility and recommendations

`pip install -r requirements.txt → python test_pipeline.py → python evaluate.py` (controlled set; writes eval_report.json) → `python evaluate.py --real` (downloads MultiWOZ shard on demand) → `python app.py` (UI/API on :8000). Offline and deterministic; generated sets use fixed seeds 1 (calibration) and 2 (test); no API key needed.

- Next steps:** independent human annotation of real conversations; a paraphrase-robust detector for the hard_pos slice; per-risk-type subgroup audit; test live LLM mode before any demo; add authentication and rate limiting before public deployment.